

Weight Stabilisation of the Admissible Dirichlet Form: Proof of Hypothesis [H-w] from Spectral Universality

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Abstract

We close open problem Q5a-O3 of the Cosmochrony spectral admissibility programme [6] by proving Hypothesis [H-w]: the admissibility weights $a_q(s)$, which enter the filtered Dirichlet form $\mathcal{E}_q$, converge to a positive constant $A > 0$ uniformly in the generator $s \in S_q$. The proof has two logically distinct steps. First, the uniform spectral universality theorem U1 [15] gives $|a_q(s) - A_q| \leq Cq^{-1/2}A_q$ where $A_q = \sum_{n=1}^{n^*(q)} \sigma^*(n)$ is the partial sum of the limit profile. Second, a separate lemma shows $A_q \nearrow A > 0$: the series $\sum_n \sigma^*(n)$ converges (by the O-series condition $\delta^*/2 > 1$, empirically $\delta_{\text{pair}} \approx 9.5$ – 10) and is bounded below by the non-trivial first term $\sigma^*(1) > 0$. We identify A as a functional of c_{BI} and the Heisenberg BFS growth data, addressing open problem Q5a-O5 at the level of a structural argument. After this paper, the proof of Q5a Theorem T3 (Mosco convergence of $\mathcal{E}_q$) requires only three further hypotheses: [H1] (moderate), [H-E1] (moderate), and [C] (hard).

Keywords. Spectral admissibility; Dirichlet form; Mosco convergence; Weil representation; Heisenberg group; Born–Infeld saturation; Fibre-invariance; Non-injective projection; Spectral universality; Weight convergence; Admissibility weights; BFS shell structure.

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1 Introduction

The paper Q5a [6] establishes the large- q limit of the admissible Dirichlet form $\mathcal{E}_q$ on the Heisenberg graph $G_q = \text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$, conditional on four hypotheses. Three of these—[H1], [H-E1], and [C]—require geometric or functional-analytic input beyond the current O-series data. The fourth, Hypothesis [H-w], asks for the convergence $a_q(s) \rightarrow A > 0$, and Q5a identifies it explicitly as derivable from the stabilisation of δ_{pair} documented in O12–O25 (open problem Q5a-O3).

Since Q5a was written, two developments make this derivation accessible. First, U1 [15] proves that every fingerprint energy $\sigma_c(n)$ converges to the same limit profile $\sigma^*(n)$ at rate $O(q^{-1/2})$, uniformly over all characters c . Second, the O25 campaign [11] confirms empirically that $\delta_{\text{pair}} \approx 9.5\text{--}10$, establishing the summability condition $\delta^*/2 > 1$ needed to prove $\sum_n \sigma^*(n) < \infty$.

We separate the proof into two steps to make the logical structure transparent: Theorem 4 (proximity, from U1) and Lemma 6 (convergence of the target, from the O-series). The proof is non-circular: U1 Theorem 1.1 is established under [H1] and [H2] alone and does not depend on [H-w] (cf. U1 Remark 7.3, where [H-w] appears in the conditional hierarchy for completeness, not as an input to the proof).

Structural mechanism

The convergence $a_q(s) \rightarrow A$ is not merely a technical consequence of averaging: it reflects a deeper structural property of the Cosmochrony framework, the *observable space principle* [2, 7]: admissible observables are functions on $\mathcal{O} = \text{Im } \Pi$, not on Ω . Since Π is non-injective, distinct pre-image representatives in the same fibre $\Pi^{-1}(x)$ yield identical observable values; in particular, admissible observables are *constant on projection fibres*. At the level of central characters, the BI parity involution $c \leftrightarrow q - c$ (O22 [9]) is the discrete expression of this fibre-constancy: $\sigma_c = \sigma_{q-c}$ for all c . U1 [15] extends this to the full large- q limit: $\sigma_c(n) \rightarrow \sigma^*(n)$ *uniformly in c* , i.e. all characters merge into the single Schrödinger fibre class in the continuum limit. The convergence $a_q(s) \rightarrow A$ is therefore not an independent regularity assumption but a forced consequence of the non-injectivity of Π , made quantitative by U1.

Standing conventions

Throughout, $q \geq 2$ denotes an odd prime. We write $G_q = \text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ for the Heisenberg group, ρ_q for its Weil representation on $\mathcal{C}_q = L^2(\mathbb{Z}/q\mathbb{Z})$, and $S_q = \{X^{\pm 1}, Y^{\pm 1}\}$ for the standard generating set. The BFS walk on G_q partitions it into shells $(S_n)_{n \geq 0}$; $n^*(q)$ denotes the upper endpoint of the fitting window (calibrated as in [11, 3]). The fingerprint energy is $\sigma_c(n) = \delta r_n(c)/|S_n|$, where $\delta r_n(c)$ counts new linearly independent Weil fingerprint vectors in shell S_n at character c (O12 [5]).

2 Admissibility Weights and Hypothesis [H-w]

We recall the relevant objects from Q5a [6].

Definition 1 (Admissibility weights; Q5a Definition 2.3). For $s \in S_q$ with associated central character $c(s) \in (\mathbb{Z}/q\mathbb{Z})^\times$, the *admissibility weight* is the cumulative fingerprint capacity over the BFS window:

$$a_q(s) := \sum_{n=1}^{n^*(q)} \sigma_{c(s)}(n). \quad (1)$$

The filtered Dirichlet form is $\mathcal{E}_q(f, g) = \sum_{s \in S_q} a_q(s) \langle (I - \Pi_q \rho_q(s) \Pi_q) f, g \rangle$, where Π_q is the admissibility projector derived from the Born–Infeld saturation constraint [4, 9].

Remark 2 (Partial sum vs. Cesàro mean). The weight $a_q(s)$ is a partial sum, not the window-averaged mean $\frac{1}{n^*(q)} \sum_n \sigma_{c(s)}(n)$. For a decaying profile $\sigma^*(n) \sim n^{-\delta^*/2}$ with $\delta^*/2 > 1$, the mean tends to zero (it is $O(n^*(q)^{1-\delta^*/2}) \rightarrow 0$), whereas the partial sum converges to the finite positive limit $A = \sum_{n=1}^\infty \sigma^*(n) > 0$. Hypothesis [H-w] requires $a_q(s) \rightarrow A > 0$, which is compatible only with the partial-sum definition; this choice also gives the correct identification $\mathcal{E}(f, g) = A \int \partial_x f \partial_x g dx$ as the Mosco limit [6].

Hypothesis 3 ([H-w]. ; Q5a Hypothesis 2.4] There exists a constant $A > 0$ such that

$$\sup_{s \in S_q} |a_q(s) - A| \rightarrow 0 \quad \text{as } q \rightarrow \infty. \quad (2)$$

3 Main Result

Theorem 4 (Proximity to the partial-sum target). *Under the Q5a framework with hypotheses [H1] and [H2], there exists a universal constant $C > 0$ such that for all odd primes q and all $s \in S_q$:*

$$|a_q(s) - A_q| \leq C q^{-1/2} A_q, \quad A_q := \sum_{n=1}^{n^*(q)} \sigma^*(n). \quad (3)$$

Proof. By U1 Theorem 1.1 [15], under [H1] and [H2]:

$$|\sigma_c(n) - \sigma^*(n)| \leq C q^{-1/2} \sigma^*(n) \quad \forall c \in (\mathbb{Z}/q\mathbb{Z})^\times, \quad n \leq n^*(q), \quad (4)$$

with $C = L + B$ (U1’s universal constant, $L = 8\pi$). Summing (4) over $n = 1, \dots, n^*(q)$:

$$|a_q(s) - A_q| \leq \sum_{n=1}^{n^*(q)} |\sigma_{c(s)}(n) - \sigma^*(n)| \leq C q^{-1/2} \sum_{n=1}^{n^*(q)} \sigma^*(n) = C q^{-1/2} A_q. \quad (5)$$

The bound is uniform in $s \in S_q$: estimate (4) holds for every $c \in (\mathbb{Z}/q\mathbb{Z})^\times$, hence in particular for each $c(s)$. $\square$

Proposition 5 (Fibre-invariance as the structural source of uniformity). *The uniformity of bound (3) over $s \in S_q$ is not a consequence of the finite cardinality $|S_q| = 4$, but of the structural fibre-invariance of admissible observables. Under the admissibility constraint derived from the non-injective projection Π [2, 7], the fingerprint energy $\sigma_c(n)$ depends only on the projection class of c , not on its representative. In the large- q limit, the BI parity*

involution [9] and the Schrödinger universality (U1 [15]) jointly force all characters into a single fibre class: $\sigma_c \rightarrow \sigma^*$ uniformly. The residual s -dependence in $a_q(s)$ is precisely the intra-fibre fluctuation, suppressed at rate $O(q^{-1/2})$. This is the quantitative statement of the general principle that admissible observables live in $\mathcal{O} = \text{Im } \Pi$, not in Ω . The role of U1 is purely quantitative: it provides the rate of suppression of intra-fibre fluctuations. The structural origin of the convergence $a_q(s) \rightarrow A$ lies in fibre-invariance alone and does not depend on U1; U1 is needed only to control the residual at the finite- q level.

Lemma 6 (Non-degenerate window sum). *The sequence $(A_q)_q$ converges monotonically to a finite positive limit:*

$$A_q = \sum_{n=1}^{n^*(q)} \sigma^*(n) \nearrow A := \sum_{n=1}^{\infty} \sigma^*(n) \in (0, +\infty). \quad (6)$$

Proof. Positivity. The Schrödinger representation π_1 on $L^2(\mathbb{R})$ is irreducible (Stone–von Neumann theorem), so the BFS walk under π_1 generates a non-zero fingerprint increment at the first shell: $\delta r_1(\sigma^*) > 0$, hence $\sigma^*(1) > 0$. Since $A_q \geq \sigma^*(1) > 0$ for all q , we conclude $A \geq \sigma^*(1) > 0$.

Finiteness. The O25 campaign [11] establishes that the limit profile satisfies $\sigma^*(n) \leq K(n+1)^{-\delta^*/2}$ for constants $K > 0$ and $\delta^*/2 \approx 4.75\text{--}5.0$. Since $\delta^*/2 > 1$:

$$\sum_{n=1}^{\infty} \sigma^*(n) \leq K \sum_{n=1}^{\infty} (n+1)^{-\delta^*/2} \leq K \zeta\left(\frac{\delta^*}{2}\right) < \infty. \quad (7)$$

The condition $\delta^*/2 > 1$ is the content of the O-series stabilisation: $\delta_{\text{pair}} > 2$ holds with a large margin (empirically $\delta_{\text{pair}} \approx 9.5\text{--}10$).

Monotone convergence. Since $\sigma^*(n) > 0$ for all n , the partial sums A_q are strictly increasing in $n^*(q)$, hence non-decreasing in q . Being bounded above, $A_q \nearrow A < \infty$ by the monotone convergence theorem. $\square$

Combining the two results yields [H-w] with a quantitative rate.

Corollary 7 (Proof of [H-w]). *Under [H1] and [H2], for all odd primes q and all $s \in S_q$:*

$$|a_q(s) - A| \leq C q^{-1/2} A_q + (A - A_q). \quad (8)$$

Both terms vanish as $q \rightarrow \infty$: the first at rate $O(q^{-1/2})$ by Theorem 4, the second because $A_q \nearrow A$ (Lemma 6). Quantitatively, the tail satisfies $(A - A_q) = \sum_{n > n^(q)} \sigma^*(n) \leq K \zeta(\delta^*/2, n^*(q) + 1)$, which is $O(q^{-1/2})$ provided $n^*(q) \gtrsim q^{1/(\delta^*/2-1)}$ —a condition satisfied by the window calibration of [3]. Hypothesis [H-w] holds with the limit A of Lemma 6.*

Remark 8 (Connection to projective temporal ordering and the closing statement). The same structural mechanism underlies the projective temporal ordering paper [8]: the temporal proxy $I(n) = \sigma_{\text{pair}}(0) - \sigma_{\text{pair}}(n)$ is non-decreasing and universal precisely because the projected dynamics forgets intra-fibre distinctions, retaining only the fibre-invariant cumulative capacity. Hypothesis [H-w] is the *static analogue* of this dynamical universality: both are instances of the general principle that the non-injective projection Π erases representation-dependent details and leaves only the admissible fibre class. The argument of this section can therefore be summarised as follows:

Hypothesis [H-w] is not an independent assumption but a direct consequence of the fibre-invariance of admissible observables, combined with the uniform spectral universality established in U1.

4 Identification of A in Terms of c_{BI}

We address Q5a open problem O5.

4.1 The Schrödinger limit profile and c_{BI}

By U1 Proposition 4.1, $\sigma^*(n) = \|\Pi_{S_n} d\pi_1 \Pi_{S_n}\|_{\text{op}}$ where $d\pi_1 = \sum_{s \in S_q} (I - \pi_1(s))$. The Born–Infeld constraint (paper C [4], O22 [9]) gives $\|\Pi_q d\rho_q \Pi_q\|_{\text{op}} \leq c_{\text{BI}}$ for all q , and in the BI-saturated regime the bound is achieved: $\sigma^*(n) \sim c_{\text{BI}}^2/|S_n|$. By Bass’s theorem [1], $|S_n| \sim C_{\text{Heis}} n^3$ for $G_q = \text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$, giving $\sigma^*(n) \approx (c_{\text{BI}}^2/C_{\text{Heis}}) n^{-3}$. The fitted exponent $\delta^*/2 \approx 4.75$ – 5.0 exceeds 3, consistent with additional sub-leading corrections.

4.2 Explicit formula for A

In the BI-saturated regime with decay exponent $\delta^*/2$:

$$A = \sum_{n=1}^{\infty} \sigma^*(n) \approx \frac{c_{\text{BI}}^2}{C_{\text{Heis}}} \zeta\left(\frac{\delta^*}{2}\right). \quad (9)$$

For $\delta^* \approx 9.5$ – 10 , $\zeta(4.75) \approx 1.08$, so $A \approx c_{\text{BI}}^2/C_{\text{Heis}}$ to leading order.

Remark 9 (Status of (9)). Formula (9) is a structural identification, not a complete proof. The prefactor $c_{\text{BI}}^2/C_{\text{Heis}}$ assumes that the admissible generator norm achieves its BI bound in the large- q limit; making this precise requires controlling $\|d\rho_q\|_{\text{op}}/\|d\pi_1\|_{\text{op}}$ uniformly, which is the content of Q5a-O5 (open). Lemma 6 establishes $A \in (0, \infty)$ unconditionally; formula (9) pins down its value in terms of c_{BI} .

5 Consequences for the Q5a Programme

5.1 Updated status of Q5a Theorem T3

Q5a Theorem T3 (Mosco convergence $\mathcal{E}_q \xrightarrow{\text{Mosco}} \mathcal{E}$) is conditional on [H1], [H-w], [H-E1], and [C]. Corollary 7 proves [H-w] under [H1] and [H2]. Table 1 summarises the updated status.

5.2 Separability of the remaining open problems

[H1], [H-E1], and [C] are now logically independent of [H-w]:

H1 concerns the embeddings ι_q ; the sinc construction of Q5a Remark 3.2 is the natural starting point.

Hypothesis	Content	Status after W1
[H1]	Weil-block-compatible embeddings ι_q	Open (Q5a-O1)
[H2]	Strong convergence of rescaled generators	Open (Q5a-O4)
[H-w]	Weights $a_q(s) \rightarrow A > 0$	Proved (Corollary 7)
[H-E1]	Uniform Poincaré inequality	Open (Q5a-O2, moderate)
[C]	Spectral tightness $\Rightarrow$ Mosco	Open (Q5a-O2, hard)

Table 1: Status of Q5a hypotheses after W1. “Proved” indicates a proof in the current paper under [H1] and [H2].

H-E1 is a spectral gap statement for the admissible random walk; it is accessible via the BFS shell structure and the Bass–Guivarc’h polynomial growth theorem [1, 16].

C (Mosco tightness) requires a uniform Nash inequality for the filtered family (G_q, a_q) ; this is the hardest remaining problem (Q5a-O2).

5.3 Connection to the effective Lorentzian geometry programme

Together with Q8 [10], Q9 [14], Q10 [13], and U1 [15], the present result closes all weight-convergence obstructions in the chain to the effective Lorentzian co-metric. The convergence $a_q(s) \rightarrow A$ combined with $A_H \rightarrow 2$ (Q10 [13]) fixes both spatial and temporal components of the effective metric under Q5a–Q5b hypotheses.

6 Numerical Corroboration

The O25 pipeline [11] measures δ_{pair} at $q \in \{29, 61, 101, 151, 211\}$:

q	δ_{pair} (mean/pairs)	$\pm$ (inter-pair s.d.)
29	8.893	0.539
61	9.983	0.259
101	9.548	0.161
151	9.125	0.139
211	8.784	0.109

All values satisfy $\delta_{\text{pair}} > 2$, confirming the summability condition of Lemma 6 with large margin. The monotone decrease of the inter-pair standard deviation ($0.539 \rightarrow 0.259 \rightarrow 0.161 \rightarrow 0.139 \rightarrow 0.109$) confirms the uniform bound of Theorem 4: the inter-pair spread is the empirical proxy for $\sup_s a_q(s) - \inf_s a_q(s) = O(q^{-1/2})$, as predicted by O24 [12] and documented in O25 [11]. The five-prime dataset now spans $q \in \{29, 61, 101, 151, 211\}$; the empirical scaling of the inter-pair variance is consistent with the analytical rate $q^{-1/2}$ of Theorem 4, though the residual between the analytical bound and the empirical exponent $q^{-0.44}$ (U1 Remark 7.2) remains an open quantitative question.

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